ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. CSE (AI)

Semester: II

SUBJECT CODE: BCSAI201

DISCRETE MATHEMATICS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

(i) Write a short note on preposition.

(ii) What is the difference between relation and function?

(iii) Convert the following statement into symbolic form: *"If it rains, then the ground is wet. If the ground is wet, then the match is postponed."* Is this a tautology?

(iv) Define planer graph with example.

(v) Define complete and regular graph.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What do you mean by power set?

(ii) State De Morgan's law and Absorption Law.

(iii) Define group.

(iv) Write associative and distributive law in preposition logic.

(v) Describe abelian group in sort.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Identify whether the each of the following relations defined on the set X ={1,2,3,4} are reflexive, symmetric, transitive and/or anti-symmetric?

- R_1 = { (1,1), (1,2), (2,1) }
- R_2 = { (1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4) }
- R_3 = { (2,1), (3,1), (3,2), (4,1), (4,2), (4,3) }

OR

(b) Let A={1,2,3,4,5}, B={3,4,5,6,7}. Find: A∩B, A∪B, and A−B.

(ii)(a) If G={1,2,3,4,5,6} under multiplication modulo 7, verify whether it forms a cyclic group.

OR

(b) Show that the statements **P↔Q** and **(P ∧ Q) V(¬ P ∧ ¬ Q)** are equivalent

Q.4.(i)(a) Express the following

- Euler graph and Hamiltonian graph
- Walk and path
- Bipartite graph

OR

(b) If a set contains 8 elements, how many subsets does it has? Explain in brief.

(ii)(a) Identify whether **(p Λ q) → (p V q)** is tautology or contradiction with using Truth table.?

OR

(b) Define the binary operation * on Z by x*y=x + y + 1 for all x, y belongs to set of integers. Verify that (Z,*) is abelian group?

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5. (a) Let **G = {1,-1**, i, —i**}** with the operation of ordinary multiplication on G be an algebraic structure, where $i=\sqrt{-1}$.

- Determine whether G is abelian.
- Determine the order of each element in G.
- Determine a subgroup of the group G.

OR

(b) Let (G,*) and G',*') be any two groups and let e and e' be their respective identities. If f is a homomorphism of G into G', then prove that

- f(e) = e'
- $f(x^{-1}) = [f(x)]^{-1}$, ∀ x ∈G

Q.6. (a) Solve the recurrence relation using generating function.

a_{n+2}- 5a_{n+1} +6a_n =2, with a_0=3 and a_1=7.

OR

(b) Construct the truth table for the following statements:

i) (P→Q')→P' ii) P↔(P'∨Q').